RESISTANCE-REGULARITY UNDER GRAPH JOINS*

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Abstract. Zhou, Wang and Bu introduced resistance-regular graphs, namely connected graphs whose resistance distance matrix has constant row sum, and proved several equivalent criteria for this property. This note applies their diagonal criterion to graph joins. Inverse formulae for complete joins of connected graphs and resistance-distance formulae for more general H-joins are already available in the literature; the purpose here is not to claim a new general inverse formula. Rather, we record the join inverse in the following shifted-resolvent form, extending the connected-factor formula to arbitrary nonempty, possibly disconnected, factors, and use its diagonal part for resistance-regularity. For graphs G and H of orders m and n, with $N = m + n$, the formula is

$$L^+_{G\vee H} = \begin{pmatrix} (L_G + nI_m)^{-1} + \left(\dfrac{n}{mN^2} - \dfrac{1}{mn}\right) J_m & -\dfrac{1}{N^2} J_{m,n} \\ -\dfrac{1}{N^2} J_{n,m} & (L_H + mI_n)^{-1} + \left(\dfrac{m}{nN^2} - \dfrac{1}{mn}\right) J_n \end{pmatrix}.$$

Consequently, $G\vee H$ is resistance-regular precisely when the two shifted Laplacian resolvent diagonals are constant and satisfy one scalar balance condition. We apply this criterion to prove that $K_m \vee H$ is resistance-regular if and only if H is complete, so the only resistance-regular cones are complete graphs. We also record, as a direct consequence of known complete multipartite resistance/inverse formulae, that complete multipartite graphs are resistance-regular exactly in the balanced case, and we give a scalar spectral criterion for joins of walk-regular factors. In particular, $X \vee X$ is resistance-regular for every connected walk-regular graph X, while joins of two resistance-regular factors need not be resistance-regular.

1. Introduction

Throughout the paper, graphs are finite, simple and undirected. A graph is assumed to be connected whenever its resistance matrix is discussed. If every edge of a connected graph G is replaced by a unit resistor, the *resistance distance* r_{uv} between two vertices $u, v \in V(G)$ is the effective electrical resistance between them. The matrix

$$R_G = (r_{uv})_{u,v\in V(G)}$$

is the *resistance matrix* of G. Following Zhou, Wang and Bu [13], a connected graph G is called *resistance-regular* if all row sums of R_G are equal.

The basic recognition tools for resistance-regularity are already known. Zhou, Wang and Bu proved that resistance-regularity is equivalent to the constancy of the diagonal of the Moore–Penrose inverse L^+_G of the Laplacian, and also to the constancy of the diagonal of the ordinary inverse $(L_G+|V(G)|^{-1}J)^{-1}$ [13, Theorem 12]. Sarma recorded the row-sum identity and the equivalent Kirchhoff-index and eigenvector forms [10, Theorem 2.5, Corollary 2.6, Theorem 2.29]. The local Foster-type condition is also contained in [13, Theorem 12] and appears in [10, Lemma 2.23]. A resistance-curvature formulation was developed in [4, 11].

The aim of this note is not to reprove these criteria, but to use the diagonal criterion to study graph joins. For two nonempty graphs G and H, the join $G \vee H$ is obtained from

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their disjoint union by adding all edges between $V(G)$ and $V(H)$. Inverse formulae for joins have already appeared in broader settings. In particular, Bapat, Karimi and Liu gave a Moore–Penrose inverse formula for complete joins of connected graphs [3, Proposition 3.3]. Moreover, the ordinary two-factor join is the base-graph K_2 case of the H-join construction, and resistance distances in general H-joins were studied by Zhang, Zhao, Liu and Arockiaraj through Laplacian generalized inverses and group inverses [12].

Thus the novelty here is not the existence of a general inverse formula for joins. The formula in Theorem 3.1 should be read as an attributed shifted-resolvent reformulation of known complete-join inverse formulae in the connected case, together with a short spectral derivation that also permits disconnected factors. The contribution is the extraction of the diagonal information needed for resistance-regularity and the subsequent applications. The resulting criterion reduces resistance-regularity of $G \vee H$ to constancy of two shifted Laplacian resolvent diagonals plus a scalar balance equation. This formulation gives short proofs of several consequences: a complete classification of universal-clique joins, a complete multipartite classification obtained from known complete multipartite resistance/inverse formulae, and a spectral criterion for joins of walk-regular factors.

2. Known diagonal tools

Let $G = (V, E)$ be a connected graph on n vertices. Its adjacency matrix is A, its diagonal degree matrix is D, and its Laplacian is $L = D - A$. The identity matrix is denoted by I, the all-one vector by $\mathbf{1}$, and $J = \mathbf{1}\mathbf{1}^\top$. More generally, $J_{a,b}$ denotes the $a \times b$ all-one matrix, and $J_a = J_{a,a}$. Since G is connected,

$$LL^+ = L^+L = I - \frac{1}{n}J, \qquad L^+\mathbf{1} = 0. \tag{2.1}$$

The resistance distance has the standard generalized-inverse expression

$$r_{uv} = (L^+)_{uu} + (L^+)_{vv} - 2(L^+)_{uv}, \tag{2.2}$$

see, for example, [1, 2, 7, 9]. The Kirchhoff index is

$$\mathrm{Kf}(G) = \frac{1}{2} \sum_{u,v \in V} r_{uv}.$$

If the Laplacian eigenvalues are $0 = \lambda_1 < \lambda_2 \le \cdots \le \lambda_n$, then

$$\mathrm{Kf}(G) = n \sum_{k=2}^{n} \frac{1}{\lambda_k} = n \operatorname{tr}(L^+), \tag{2.3}$$

a classical formula due to Gutman and Mohar [6].

The next statement is quoted as a known theorem. It is the basic tool used in the sequel.

Theorem 2.1 (Known diagonal criteria)**.** *Let G be a connected graph on n vertices. The following statements are equivalent.*

(1) *G is resistance-regular.*
(2) *$R_G\mathbf{1} = \rho\mathbf{1}$ for some scalar ρ.*
(3) *The diagonal entries of L_G^+ are all equal.*
(4) *For every vertex v,*
$$(L_G^+)_{vv} = \frac{\mathrm{Kf}(G)}{n^2}.$$
(5) *If $q_2, \ldots, q_n$ are orthonormal eigenvectors of L_G associated with $\lambda_2, \ldots, \lambda_n$, then*
$$\sum_{k=2}^{n} \frac{q_k(v)^2}{\lambda_k}$$
is independent of v.

When these conditions hold, the common row sum of R_G is $2\,\mathrm{Kf}(G)/n$.

The equivalence with constant diagonal of L_G^+ is contained in [13, Theorem 12]. Sarma's row-sum identity

$$R_i = \frac{\mathrm{Kf}(G)}{n} + n(L_G^+)_{ii} \tag{2.4}$$

for the i-th row sum [10, Theorem 2.5] is equivalent to

$$R_G\mathbf{1} = n\,\mathrm{diag}(L_G^+) + \mathrm{tr}(L_G^+)\mathbf{1}. \tag{2.5}$$

The $\mathrm{Kf}(G)/n^2$ and eigenvector forms are included in [10, Theorem 2.29].

3. Block and diagonal formulae for graph joins

The next theorem is the calculation used throughout the paper. For connected factors it is the two-factor specialization and shifted-resolvent rewriting of the complete-join Moore–Penrose inverse formula of Bapat, Karimi and Liu [3, Proposition 3.3]; see Remark 3.2. The spectral proof below also covers disconnected factors.

Theorem 3.1 (Shifted-resolvent form of the join inverse)**.** *Let G and H be nonempty graphs of orders m and n, respectively, and put $N = m+n$. Let L_G and L_H denote their Laplacians. With the vertex ordering $V(G), V(H)$,*

$$L_{G\vee H}^+ = \begin{pmatrix} (L_G + nI_m)^{-1} + \left(\frac{n}{mN^2} - \frac{1}{mn}\right) J_m & -\frac{1}{N^2} J_{m,n} \\ -\frac{1}{N^2} J_{n,m} & (L_H + mI_n)^{-1} + \left(\frac{m}{nN^2} - \frac{1}{mn}\right) J_n \end{pmatrix}. \tag{3.1}$$

In particular, for $x \in V(G)$ and $y \in V(H)$,

$$(L_{G\vee H}^+)_{xx} = (L_G + nI_m)^{-1}_{xx} - \frac{1}{mn} + \frac{n}{mN^2}, \tag{3.2}$$

$$(L_{G\vee H}^+)_{yy} = (L_H + mI_n)^{-1}_{yy} - \frac{1}{mn} + \frac{m}{nN^2}. \tag{3.3}$$

Consequently, $G \vee H$ is resistance-regular if and only if the diagonal of $(L_G + nI_m)^{-1}$ is constant on $V(G)$, the diagonal of $(L_H + mI_n)^{-1}$ is constant on $V(H)$, and their common diagonal values α and β satisfy

$$\alpha + \frac{n}{mN^2} = \beta + \frac{m}{nN^2}. \tag{3.4}$$

Proof. Choose orthonormal Laplacian eigenbases

$$m^{-1/2}\mathbf{1}_G, q_2, \ldots, q_m \quad \text{and} \quad n^{-1/2}\mathbf{1}_H, p_2, \ldots, p_n$$

of G and H, with eigenvalues $0, \lambda_2, \ldots, \lambda_m$ and $0, \mu_2, \ldots, \mu_n$, respectively. If a factor is disconnected, the additional zero-eigenvectors are included among the vectors q_i or p_j; in all cases these vectors are chosen orthogonal to the appropriate all-one vector. The Laplacian of $G \vee H$ has eigenvectors

$$(q_i, 0) \quad (2 \le i \le m), \qquad (0, p_j) \quad (2 \le j \le n),$$

with eigenvalues $\lambda_i + n$ and $\mu_j + m$, respectively. In addition to the all-one eigenvector, the remaining normalized eigenvector is constant on each part and can be taken as

$$z = \left(\sqrt{\frac{n}{mN}}\,\mathbf{1}_G, -\sqrt{\frac{m}{nN}}\,\mathbf{1}_H\right),$$

with eigenvalue N.

The spectral decomposition of $L_{G\vee H}^+$ is therefore

$$\sum_{i=2}^{m} \frac{1}{\lambda_i + n}(q_i, 0)(q_i, 0)^\top + \sum_{j=2}^{n} \frac{1}{\mu_j + m}(0, p_j)(0, p_j)^\top + \frac{1}{N} zz^\top.$$

Since

$$\sum_{i=2}^{m} \frac{q_i q_i^\top}{\lambda_i + n} = (L_G + nI_m)^{-1} - \frac{1}{mn} J_m,$$

and similarly for H, while $N^{-1}zz^\top$ contributes $\frac{n}{mN^2}J_m$ to the G-block, $\frac{m}{nN^2}J_n$ to the H-block, and $-N^{-2}J_{m,n}$ to each off-diagonal block, formula (3.1) follows. The diagonal formulae (3.2) and (3.3) are immediate. The final assertion follows from the known diagonal criterion in Theorem 2.1, because $G \vee H$ is connected for nonempty G and H. □

Remark 3.2 (Relation with existing join inverse formulae)*.* For connected factors, Theorem 3.1 is equivalent to the complete-join Moore–Penrose inverse formula of Bapat, Karimi and Liu [3, Proposition 3.3]. In the two-factor case their formula can be written as

$$L_{G\vee H}^{+} = -\frac{1}{N^2} J_N + \left(NI_m - L_{\overline{G}}\right)^{-1} \oplus \left(NI_n - L_{\overline{H}}\right)^{-1}. \tag{3.5}$$

Since $L_{\overline{G}} = mI_m - J_m - L_G$, one has

$$NI_m - L_{\overline{G}} = L_G + nI_m + J_m.$$

The Sherman–Morrison identity gives

$$(L_G + nI_m + J_m)^{-1} = (L_G + nI_m)^{-1} - \frac{1}{nN} J_m,$$

and the analogous identity holds for H. Substituting these identities in (3.5) gives exactly (3.1). The present derivation is included because it is short, works without connectedness assumptions on the factors, and displays the shifted-resolvent diagonal needed for the resistance-regularity criterion. The more general H-join results of Zhang, Zhao, Liu and Arockiaraj [12] provide a broader framework for resistance distances under graph compositions; here the focus is the special diagonal condition defining resistance-regularity.

4. Consequences for resistance-regular joins

4.1. **Universal-clique joins and cones.**

The join formula gives a complete classification when one factor is a complete graph.

Theorem 4.1 (Universal-clique joins)**.** *Let $m, n \geq 1$, and let H be a graph on n vertices. Then*

$$K_m \vee H \text{ is resistance-regular} \iff H = K_n.$$

Equivalently, a join of a nonempty clique with another graph is resistance-regular only in the complete-graph case.

Proof. If $H = K_n$, then $K_m \vee H = K_{m+n}$, which is resistance-regular. Conversely, assume that $K_m \vee H$ is resistance-regular. Put $N = m + n$, and let

$$0 = \mu_1 \leq \mu_2 \leq \cdots \leq \mu_n$$

be the Laplacian eigenvalues of H.

For $G = K_m$, the diagonal of $(L_G + nI_m)^{-1}$ is constant and equal to

$$\frac{1}{mn} + \frac{m-1}{mN}.$$

By (3.2), every vertex in the K_m part has diagonal entry

$$\frac{N-1}{N^2}$$

in $L_{K_m \vee H}^{+}$. Since $K_m \vee H$ is resistance-regular, all vertices in the H part have the same diagonal value. Summing the formula (3.3) over all vertices of H gives

$$\frac{1}{m} + \sum_{j=2}^{n} \frac{1}{\mu_j + m} - \frac{1}{m} + \frac{m}{N^2} = \frac{n(N-1)}{N^2}.$$

Hence

$$\sum_{j=2}^{n} \frac{1}{\mu_j + m} = \frac{n-1}{N}. \tag{4.1}$$

For every graph on n vertices, $\mu_j \le n$ for $2 \le j \le n$, because the complement Laplacian is

$$L_{\overline{H}} = nI_n - J_n - L_H$$

and is positive semidefinite on $\mathbf{1}^\perp$. Therefore

$$\frac{1}{\mu_j + m} \ge \frac{1}{n+m} = \frac{1}{N} \qquad (2 \le j \le n).$$

Equality in (4.1) forces $\mu_2 = \cdots = \mu_n = n$. Consequently $L_{\overline{H}}$ has all eigenvalues equal to 0, and so $\overline{H}$ has no edges. Thus $H = K_n$. □

Corollary 4.2 (Cones)**.** *Let H be a graph on $n \ge 1$ vertices. The cone $K_1 \vee H$ is resistance-regular if and only if $H = K_n$. Equivalently, the only resistance-regular cones are complete graphs.*

Proof. This is Theorem 4.1 with $m = 1$. □

4.2. **Empty factors and complete multipartite graphs.** Complete multipartite graphs have known resistance-distance and inverse formulae [5, 3]. The following resistance-regularity classification is an immediate consequence of those formulae together with the known diagonal criterion; we include the short diagonal proof for completeness.

Proposition 4.3 (Complete multipartite graphs, consequence of known formulae)**.** *Let $r \ge 2$ and let $p_1, \dots, p_r$ be positive integers. The complete multipartite graph $K_{p_1,\dots,p_r}$ is resistance-regular if and only if*

$$p_1 = p_2 = \cdots = p_r.$$

In particular, the complete bipartite graph $K_{m,n}$ is resistance-regular if and only if $m = n$.

Proof. Put $N = p_1 + \cdots + p_r$, and let V_i be the part of size p_i. The diagonal computation below is the specialization to empty factors of the known complete-join inverse formula; it is written out to make the resistance-regularity criterion transparent. The subspace

$$W_i = \{x \in \mathbb{R}^{V(K_{p_1,\dots,p_r})} : \operatorname{supp}(x) \subseteq V_i, \sum_{v \in V_i} x_v = 0\}$$

has dimension $p_i - 1$ and consists of Laplacian eigenvectors with eigenvalue $N - p_i$. The remaining nonzero eigenspace is the $(r-1)$-dimensional subspace of vectors that are constant on each part and orthogonal to $\mathbf{1}$; on this subspace the Laplacian eigenvalue is N.

For a vertex $v \in V_i$, the diagonal entry of the orthogonal projection onto W_i is $1 - 1/p_i$. The diagonal entry of the orthogonal projection onto the space of vectors constant on each part is $1/p_i$, and subtracting the projection onto $\operatorname{span}\{\mathbf{1}\}$ leaves $1/p_i - 1/N$. Therefore

$$(L^+)_{vv} = \frac{1 - 1/p_i}{N - p_i} + \frac{1}{N}\left(\frac{1}{p_i} - \frac{1}{N}\right) = \frac{N-1}{N(N-p_i)} - \frac{1}{N^2}. \tag{4.2}$$

The expression in (4.2) is strictly increasing as a function of p_i for $1 \le p_i < N$. Hence the diagonal of L^+ is constant if and only if all part sizes are equal. By Theorem 2.1, this is equivalent to resistance-regularity. □

4.3. **Walk-regular factors.** A graph is *walk-regular* if, for every $k \geq 0$, the diagonal of A^k is constant. Connected walk-regular graphs are resistance-regular by [13, Remark 2]. For joins, walk-regularity gives a more explicit spectral test.

Proposition 4.4 (Walk-regular join criterion). *Let G and H be connected walk-regular graphs of orders m and n, respectively. Let*

$$0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_m, \qquad 0 = \mu_1 < \mu_2 \leq \cdots \leq \mu_n$$

be their Laplacian spectra, and set $N = m + n$. Then $G \vee H$ is resistance-regular if and only if

$$\frac{1}{m}\left(\frac{1}{n} + \sum_{i=2}^{m} \frac{1}{\lambda_i + n}\right) + \frac{n}{mN^2} = \frac{1}{n}\left(\frac{1}{m} + \sum_{j=2}^{n} \frac{1}{\mu_j + m}\right) + \frac{m}{nN^2}. \tag{4.3}$$

In particular, $X \vee X$ is resistance-regular for every connected walk-regular graph X.

Proof. A walk-regular graph is regular. Since $L_G + nI_m$ is nonsingular, the Cayley–Hamilton theorem, equivalently spectral calculus, expresses $(L_G + nI_m)^{-1}$ as a polynomial in L_G. If G is d-regular, then $L_G = dI_m - A_G$; hence $(L_G + nI_m)^{-1}$ is a polynomial in A_G. Walk-regularity implies that every polynomial in A_G has constant diagonal, so the diagonal of $(L_G + nI_m)^{-1}$ is constant. Its common diagonal value is the trace average

$$\alpha = \frac{1}{m} \operatorname{tr} (L_G + nI_m)^{-1} = \frac{1}{m}\left(\frac{1}{n} + \sum_{i=2}^{m} \frac{1}{\lambda_i + n}\right).$$

Similarly, the common diagonal value of $(L_H + mI_n)^{-1}$ is

$$\beta = \frac{1}{n}\left(\frac{1}{m} + \sum_{j=2}^{n} \frac{1}{\mu_j + m}\right).$$

The criterion (4.3) is exactly the balance condition (3.4). If $G = H = X$, then $m = n$ and the two sides of (4.3) are identical, so $X \vee X$ is resistance-regular. □

The next example shows that resistance-regularity of both factors does not, by itself, guarantee resistance-regularity of the join.

Example 4.5. Both C_4 and C_5 are connected walk-regular, hence resistance-regular. However, $C_4 \vee C_5$ is not resistance-regular. Indeed, the Laplacian spectra are

$$\operatorname{Spec}_L(C_4) = \{0, 2, 2, 4\}$$

and

$$\operatorname{Spec}_L(C_5) = \left\{0, \frac{5-\sqrt{5}}{2}, \frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right\}.$$

For $m = 4, n = 5, N = 9$, the left side of (4.3) is

$$\frac{1}{4}\left(\frac{1}{5} + \frac{2}{7} + \frac{1}{9}\right) + \frac{5}{4 \cdot 9^2} = \frac{1867}{11340},$$

whereas the right side is

$$\frac{1}{5}\left(\frac{1}{4} + \frac{26}{41}\right) + \frac{4}{5 \cdot 9^2} = \frac{12401}{66420}.$$

These two rational numbers are unequal. Hence $C_4 \vee C_5$ is not resistance-regular.

5. Further context and closing remarks

The preceding sections deliberately separate quoted criteria from the join calculation and its applications. Every connected walk-regular graph is resistance-regular [13, Remark 2]; this includes connected vertex-transitive graphs, distance-regular graphs, strongly regular graphs, cycles, complete graphs, hypercubes and many Cayley graphs. Resistance-regular graphs on at least three vertices have no cut vertices by Sarma [10, Theorem 2.12], a fact also used in the constant-resistance-curvature setting [11, Theorem 2.5]. Thus the only resistance-regular trees are K_1 and K_2. Regularity alone is not sufficient; Sarma gives a smallest-order regular non-example found by computer search [10, p. 306, Fig. 1], and further regular non-examples are discussed in [8].

Complete multipartite graphs have been studied from the viewpoint of resistance distances and Kirchhoff-type invariants in [5, 3]. The classification recorded in Proposition 4.3 follows from the same known formulae and the diagonal criterion, and states that within this class resistance-regularity is exactly the balanced case. This complements Theorem 4.1: joining by a clique is rigid, while joining empty factors allows balanced complete multipartite examples beyond complete graphs.

The main point of this note is that the diagonal criterion becomes especially effective for joins. The block formula (3.1) reduces resistance-regularity of $G \vee H$ to a comparison of two shifted Laplacian resolvent diagonals. The universal-clique classification, the complete multipartite classification and the walk-regular join criterion suggest a broader problem: determine graph classes for which shifted Laplacian resolvent diagonals can be controlled explicitly, and use this to classify resistance-regular joins or more general graph compositions.

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